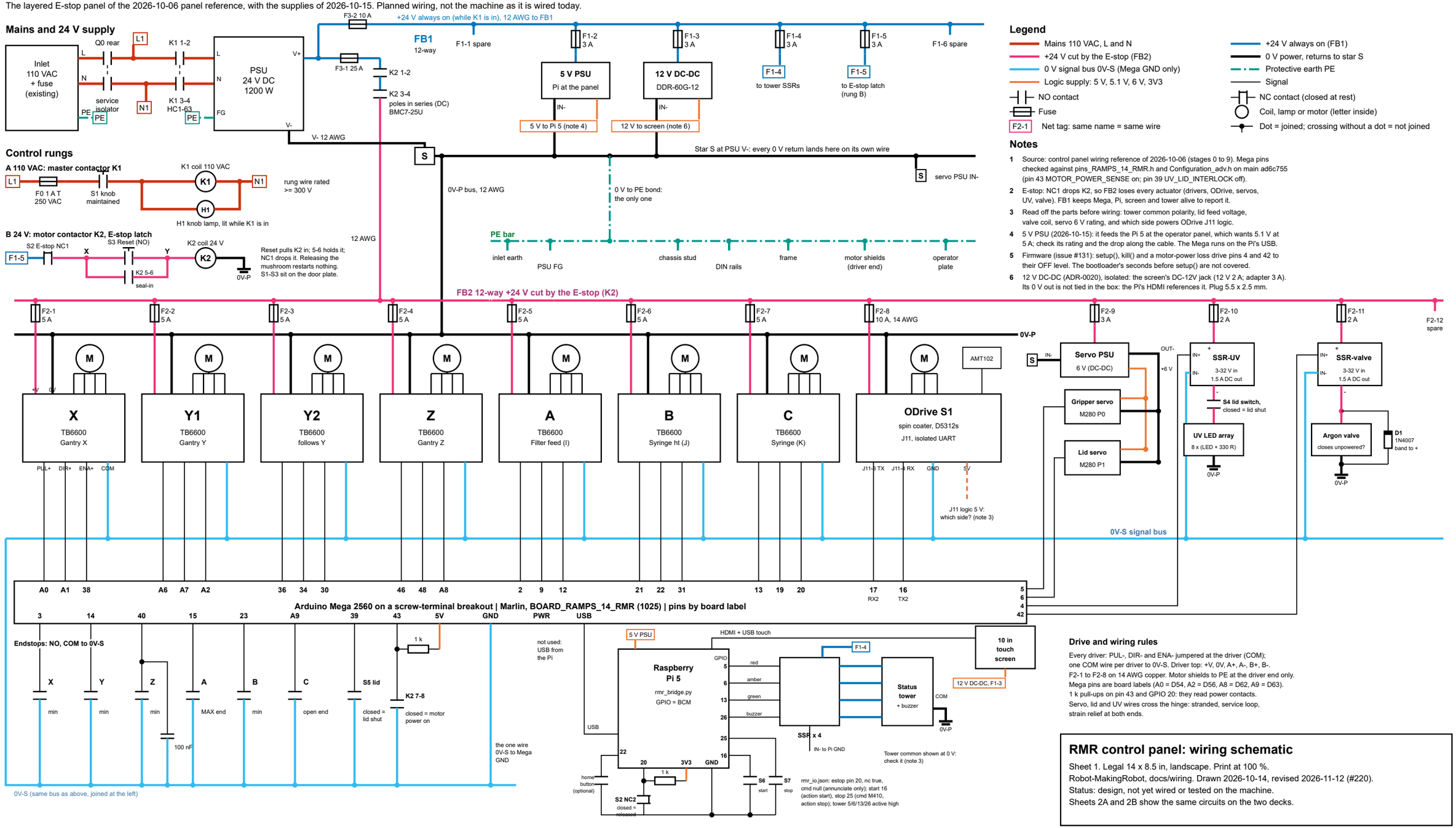
RMR control panel: wiring schematic

The layered E-stop panel of the 2026-10-06 panel reference, with the supplies of 2026-10-15. Planned wiring, not the machine as it is wired today.



Legend

- Mains 110 VAC, L and N
- +24 V cut by the E-stop (FB2)
- 0 V signal bus 0V-S (Mega GND only)
- Logic supply: 5 V, 5.1 V, 6 V, 3V3
- NO contact
- Fuse
- F2-1 Net tag: same name = same wire
- +24 V always on (FB1)
- 0 V power, returns to star S
- Protective earth PE
- Signal
- NC contact (closed at rest)
- Coil, lamp or motor (letter inside)
- Dot = joined; crossing without a dot = not joined

Notes

1. Source: control panel wiring reference of 2026-10-06 (stages 0 to 9). Mega pins checked against pins_RAMPS_14_RMR.h and Configuration_adv.h on main ad6c755 (pin 43 MOTOR_POWER_SENSE on; pin 39 UV_LID_INTERLOCK off).
2. E-stop: NC1 drops K2, so FB2 loses every actuator (drivers, ODrive, servos, UV, valve). FB1 keeps Mega, Pi, screen and tower alive to report it.
3. Read off the parts before wiring: tower common polarity, lid feed voltage, valve coil, servo 6 V rating, and which side powers ODrive J11 logic.
4. 5 V PSU (2026-10-15): it feeds the Pi 5 at the operator panel, which wants 5.1 V at 5 A; check its rating and the drop along the cable. The Mega runs on the Pi's USB.
5. Firmware (issue #131): setup(), kill() and a motor-power loss drive pins 4 and 42 to their OFF level. The bootloader's seconds before setup() are not covered.
6. 12 V DC-DC (ADR-0020), isolated: the screen's DC-12V jack (12 V 2 A; adapter 3 A). Its 0 V out is not tied in the box: the Pi's HDMI references it. Plug 5.5 x 2.5 mm.

Drive and wiring rules

Every driver: PUL-, DIR- and ENA- jumpered at the driver (COM); one COM wire per driver to 0V-S. Driver top: +V, 0V, A+, A-, B+, B-. F2-1 to F2-8 on 14 AWG copper. Motor shields to PE at the driver end only. Mega pins are board labels (A0 = D54, A2 = D56, A8 = D62, A9 = D63). 1 k pull-ups on pin 43 and GPIO 20: they read power contacts. Servo, lid and UV wires cross the hinge: stranded, service loop, strain relief at both ends.

RMR control panel: wiring schematic

Sheet 1. Legal 14 x 8.5 in, landscape. Print at 100 %.
Robot-MakingRobot, docs/wiring. Drawn 2026-10-14, revised 2026-11-12 (#220).
Status: design, not yet wired or tested on the machine.
Sheets 2A and 2B show the same circuits on the two decks.